114-3海攬班高二下重補修_學習單

Topic 1: 2D Newton's Laws & Forces

Q1-1

Force is a vector, which means it has both magnitude and direction. Imagine a box is being pulled by two ropes. Rope 1 pulls to the East with 40 N, and Rope 2 pulls to the North with 30 N.
(a) Draw these two force vectors starting from the box.
(b) Calculate the magnitude of this Net Force using the Pythagorean theorem.

- Reference Video: Visualizing vectors in 2 dimensions | Physical Processes | Khan Academy
- Explained by:
- Explanation:

Q1-2

A $10\,\mathrm{kg}$ box sits on a flat floor. The friction coefficients between the box and the floor are: static friction $\mu_s = 0.5$ and kinetic friction $\mu_k = 0.3$. (use $g = 10\,\mathrm{m/s}^2$)
(a) What is the value of the normal force ?
(b) Calculate the maximum static friction. In what situation do we use this value?
(c) Calculate the kinetic friction. In what situation do we use this value?
(d) You push the box with $40\,\mathrm{N}$ of force. Does the box move? How much friction is pushing back against you right now?
(e) Now you push with $60\,\mathrm{N}$ of force. Which friction acts on it now? What is the value of the friction force pushing back?

- Reference Video: Static and kinetic friction example | Forces and Newton's laws of motion | Khan Academy
- Explained by:
- Explanation

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Q1-3

For the following two situations, draw a dot (representing the object) and draw ALL the force vectors acting on it. Label them clearly (e.g., Gravity mg, Normal Force F_N, Pulling Force F_p, Friction f).

Situation A: A suitcase is being pulled at an angle on a rough horizontal floor.

Situation B: A block is sliding down a rough tilted ramp.

- Reference Video: Free body diagram | Forces and Newton's laws of motion | Khan Academy
- Explained by:
- Explanation:

Q1-4

You are pulling a $20\,\mathrm{kg}$ suitcase on a flat floor. Your pulling force is $50\,\mathrm{N}$ at an angle of 37° above the horizontal. the suitcase is moving , and the kinetic friction coefficient is $\mu_k = 0.1$.

(a) Calculate the horizontal pull (F_x) and the vertical pull (F_y).

(b) Calculate the normal force from the ground.

(c) Focus on y-directoin, is the normal force equal to the gravity in this case? Why or why not?

(d) Focus on x-directoin, calculate the friction force.

(e) Calculate the acceleration of the suitcase.

- Reference Video:
  Components of a force | Forces and Newton's laws of motion | Khan Academy
  Normal force and contact force | Forces and Newton's laws of motion | Khan Academy
- Explained by:
- Explanation:

Q1-5

A 10 kg block is placed on a ramp tilted at 37°. The kinetic friction coefficient is $\mu_k = 0.25$.

(a) Break Gravity into two parts: the force parallel to the ramp and the force perpendicular to the ramp. Determine them respectively.

(b)What is the value of the normal force?

(c) Calculate the kinetic friction force acting on the block.

(d) Focus ONLY on the direction parallel to the ramp. Calculate the net force

(e) Calculate the acceleration of the block sliding down the ramp.

- Reference Video:
  Inclined plane force components | Forces and Newton's laws of motion | Khan Academy
  Ice accelerating down an incline | Forces and Newton's laws of motion | Khan Academy
- Explained by:
- Explanation:

Q1-6

Imagine you slowly increase the angle of the ramp from 37° to 80° (making it very steep).
(a) Does the normal force increase or decrease? Explain your reason.
(b) Because the Normal Force changed, does the friction force between the block and the ramp become stronger or weaker?
(c) Based on your answers, why is it much harder to walk up a very steep hill without slipping?

- Reference Video: Ice accelerating down an incline | Forces and Newton's laws of motion | Khan Academy
- Explained by:
- Explanation:

Topic 2: Circular Motion

💡 Formula for centripetal acceleration

$$a_c = \frac{v^2}{R} = R\omega^2 = v\omega = \frac{4\pi^2 R}{T^2} = 4\pi^2 R f^2$$

Q2-1

Imagine a flat playground spinner as the figure shown. The distance between the center and the edge of the spinner is $2\,\mathrm{m}$. It is a perfectly flat, solid circular platform that spins horizontally without going up and down.
(a) Using the golden rule, if the spinner rotates 90°, how many radians is that?
(b) You are sitting at the edge of the spinner, and it rotates through an angle of $\theta = 6.28\,\mathrm{rad}$. Calculate the actual distance you traveled.
(c) If it takes $t = 3\,\mathrm{s}$ to complete this rotation, calculate your linear speed.

Imagine a flat playground spinner. The distance between the center and the edge of the spinner is $r = 2\,\mathrm{m}$. It is a perfectly flat, solid circular platform that spins horizontally without going up and down
(a) If the spinner rotates 90∘, how many radians is that?
(b) You are sitting at the edge of the spinner, and it rotates through an angle of $\theta = 6\,\mathrm{rad}$**.** Calculate the actual distance you traveled.
(c) If it takes $t = 3\,\mathrm{s}$ to complete this rotation, calculate your linear speed.

- Reference Video: Angular motion variables | Centripetal force and gravitation | Khan Academy
- Explained by: Nivi
- Explanation:

Q2-2

Imagine you place a small tennis ball on the very edge of the same spinner.
(a) Using the numbers from Q4-1 ($\theta = 6\,\mathrm{rad}$, $t = 3\,\mathrm{s}$), calculate the angular velocity ω of the spinner.
(b) Now, use the formula $v = r\omega$ to calculate the ball's linear speed.
(c) Draw a top-down view of the spinner. Pick a point for the ball and draw its velocity arrow. Suddenly, the ball slips! Does it fly straight outwards from the center, or in a straight line tangent to the circle? Explain using Newton's 1st Law.

- Reference Video: Centripetal acceleration | Centripetal force and gravitation | Khan Academy
- Explained by: Sarah
- Explanation:

Q2-3

Take the same situation as Q2-1. To keep the ball moving in a circle (before it slips), its velocity direction must keep changing. This requires an acceleration pointing toward the center, called centripetal

(a) Calculate the centripetal acceleration (ac) of the ball while it was on the spinner.

(b) Which direction does this acceleration point to keep its velocity changing?

- Reference Video: Centripetal acceleration | Centripetal force and gravitation | Khan Academy
- Explained by: Sarah
- Explanation:

Q2-4

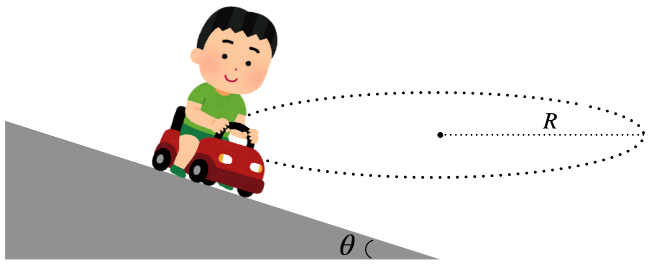
Imagine a 1000kg car turning on a smooth, frictionless road that is banked (tilted) at an angle of $\theta = 37^{\circ}$. The car is making a HORIZONTAL circle with radius R .

(a) Draw FBD of the car. Your FBD must includes gravity and normal force.

(b) Use the vertical force analysis to calculate the exact value of the normal force.

(c) Are the horizontal force balanced? Is it reasonable? Why?

(d) Calculate the centripetal force(F_c) and centripetal acceleration(a_c) of the car.



- Reference Video: Ideal banked curves | Centripetal force and gravitation | Khan Academy
- Explained by: Po
- Explanation:

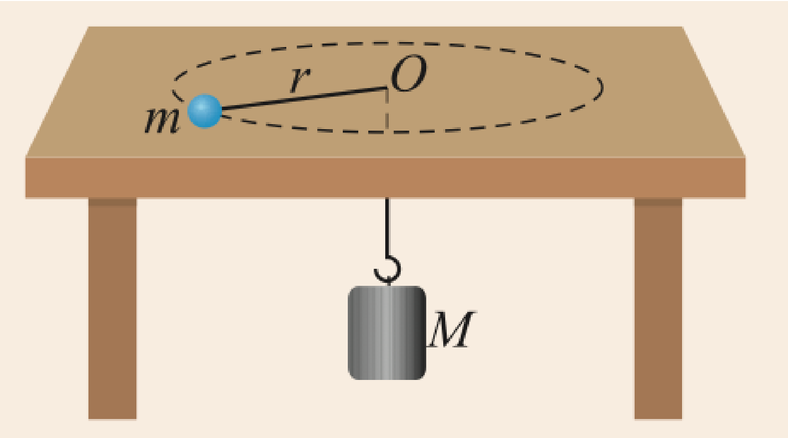
Q2-5

A small blue ball with mass $m = 2\,\mathrm{kg}$ is on a frictionless horizontal table. It is tied to a string that passes through a hole in the center (O) to a hanging weight with mass $M = 5\,\mathrm{kg}$. The blue ball is swinging in a horizontal circle with a radius of $r = 4\,\mathrm{m}$. (Use $g = 10\,\mathrm{m/s}^2$)

(a) Calculate the gravity pulling down on M, and the value of the Tension (T) in the string.

(b) Calculate the centripetal force(F_C) and centripetal acceleration(a_C) of the ball.

(c) Calculate the ball's tangential speed.



- Reference Video: Mass swinging in a horizontal circle | Centripetal force and gravitation | Khan Academy
- Explained by: Po
- Explanation:

Q2-6

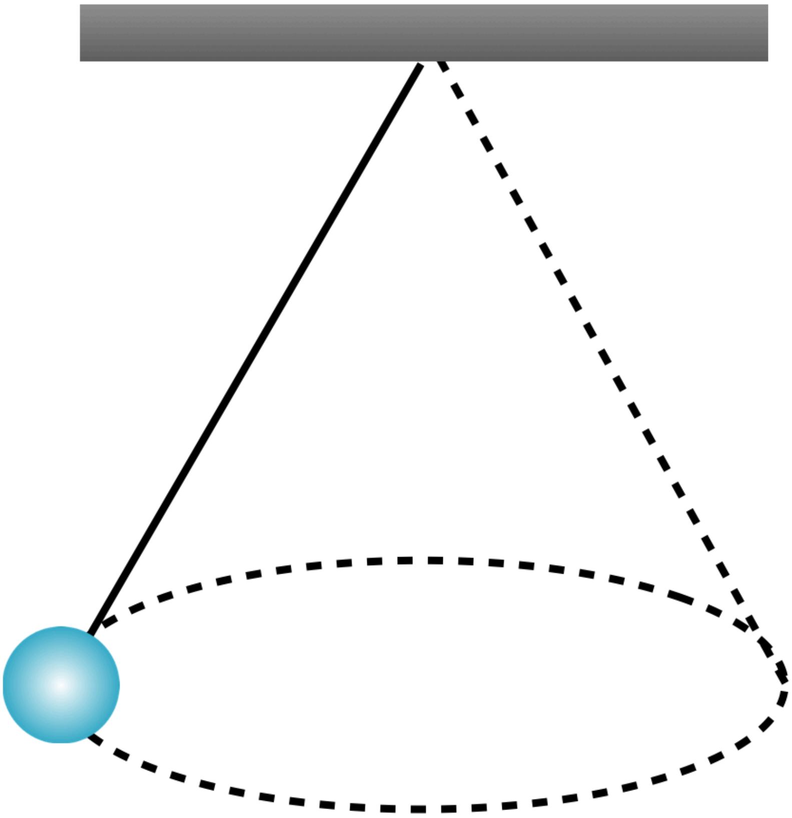
A ball with a mass of $m = 4\,\mathrm{kg}$ is flying in a horizontal circle, suspended by a string of length $L = 2\,\mathrm{m}$ attached to the ceiling. This creates a "cone" shape as it swings. The string makes an angle of 37° with the vertical straight line.(Use $g = 10\,\mathrm{m/s}^2$).

(a) Draw the FBD of the ball, and denote the centripetal force.

(b) Calculate the exact value of the tension.

(c) Calculate the radius R.

(d) Calculate the tangential speed of the ball.



- Reference Video: Centripetal force problem solving | Centripetal force and gravitation | Khan Academy (Note: The core vector breakdown concept here is universally covered across KA's centripetal force examples).
- Explained by: Nivi
- Explanation: